Assessment 2 Information

Subject Code:	TEC206
Subject Name:	Intermediate Programming
Assessment Title:	Tasks
Assessment Type:	Individual
Word Count:	Not applicable - Words (+/-10%)
Weighting:	40 %
Total Marks:	40
Submission:	MyKBS
Due Date:	Week 9

Your Task

This assessment is to be completed individually. In this assessment, you will create an application via object-oriented programming and apply best practices and design patterns to an application which meets the specifications described in the case study.

Assessment Description

This assessment is designed to test your understanding and practical application of classes and inheritance in Python 3. You will be developing a Bookstore Inventory Management System, implementing classes for different types of books.

You will be provided with a case study which will require you to use your problem-solving skills and you will create an application via object-oriented programming and apply best practices and design patterns to an application. You will need to create a solution and develop the program using Python 3 language.

Case Study:

You have been hired as a junior software developer to create a Bookstore Inventory Management System for a local bookstore. The system should help manage the inventory of books and provide essential information for stock control. You need to create classes for different types of books that inherit from a parent Book class. Each class will have specific properties, getters, setters, and methods as outlined below:

Book (class)

- Properties: `title` (string), `author` (string), `price` (float), `quantity` (integer), and `type` (one of two strings: ‘FictionBook’ or ‘NonFictionBook’)
- Getters: `getTitle`(), `getAuthor`(), `getPrice`(), `getQuantity`(), `getType`()
- Setters: `setPrice`(newPrice), `setQuantity`(newQuantity)
- Methods: A `__repr__` method that displays information about the book

FictionBook (class)

- Includes everything in the Book class, plus one additional property
- Property: `genre`(string)
- Getter: `getGenre`()
- Methods: A `__repr__` method that displays information about the book and genre

NonFictionBook (class)

- Includes everything in the Book class, plus one additional property
- Property: `subject`(string)
- Getter: `getSubject`()
- Methods: A `__repr__` method that displays information about the book and subject

NOTE:
Your program must have user interactivity, user input data validation and ease of use for user. User must be able to use the program for as long as they want.

This assessment aims to achieve the following subject learning outcomes:

LO3	Create an application via object-oriented programming.
LO4	Apply best practices and design patterns to an application.

Assessment Instructions

Assessment instructions for this assessment:

1. Read the case study provided and interpret the program specifications.
2. Develop a Python 3 program that meets the specifications outlined in the case study.
3. Submit your Python 3 program code to the designated submission platform by the due date.
    a) **You must submit your Python 3 program code in .py format extension. Any other formats will not be accepted.**
4. Implement the Python classes and methods based on the provided requirements.
5. Utilise appropriate inheritance and class properties.
6. Apply basic principles of object-oriented programming, including encapsulation and inheritance.
7. Follow the PEP 8 coding conventions and use meaningful names for variables and functions.
8. Include comments to explain the functionality and any assumptions made.
9. Test the classes with suitable inputs to ensure correctness.
10. Please refer to the assessment marking guide to assist you in completing all the assessment criteria.

Note: Aim to demonstrate your understanding of basic object-oriented programming concepts and their application. Consider code reusability and maintainability.

Important Study Information

Academic Integrity and Conduct Policy

https://www.kbs.edu.au/admissions/forms-and-policies

KBS values academic integrity. All students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the Academic Integrity and Conduct Policy.

Please read the policy to learn the answers to these questions:

- What is academic integrity and misconduct?
- What are the penalties for academic misconduct?
- How can I appeal my grade?

Late submission of assignments (within the Assessment Policy)

https://www.kbs.edu.au/admissions/forms-and-policies

Number of days late	Penalty
1* - 9 days	5% per day for each calendar day late deducted from the total marks available.
10 - 14 days	50% deducted from the total marks available.
After 14 days	Assignments submitted more than 14 calendar days after the due date will not be accepted and the student will receive a mark of zero for the assignment(s) unless special consideration, reasonable adjustment or an alternative factor related to compassionate circumstances is approved and applied.

**Assignments submitted at any stage within the first 24 hours after the deadline will be considered to be one day late and therefore subject to the associated penalty.*

Length Limits for Assessments

Penalties may be applied for assessment submissions that exceed prescribed limits.

Study Assistance

Students may seek study assistance from their local Academic Learning Advisor or refer to the resources on the MyKBS Academic Success Centre page. Further details can be accessed at https://elearning.kbs.edu.au/course/view.php?id=1481

Generative AI Traffic Lights

Please see the level of Generative AI that this assessment is Level 2 has been designed to accept:

Traffic Light	Amount of Generative Artificial Intelligence (GenerativeAI) usage	Evidence Required	This assessment (✓)
Level 1	**Prohibited:** **No GenerativeAI allowed** This assessment showcases your individual knowledge, skills and/or personal experiences in the absence of Generative AI support.	The use of generative AI is **prohibited** for this assessment and may potentially result in penalties for academic misconduct, including but not limited to a mark of zero for the assessment.	
Level 2	**Optional:** **You may use GenerativeAI for research and content generation that is appropriately referenced.** ***See assessment instructions for details*** This assessment allows you to engage with Generative AI as a means of expanding your understanding, creativity, and idea generation in the research phase of your assessment and to produce content that enhances your assessment. I.e., images. **You do not have to use it.**	The use of GenAI is **optional** for this assessment. Your collaboration with GenerativeAI **must be clearly referenced** just as you would reference any other resource type used. **Click on the link below to learn how to reference GenerativeAI.** **https://library.kaplan.edu.au/referencing-other-sources/referencing-other-sources-generative-ai** In addition, you **must** include an appendix that documents your GenerativeAI collaboration including all prompts and responses used for the assessment. Unapproved use of generative AI as per assessment details during the content generation parts of your assessment may potentially result in penalties for academic misconduct, including but not limited to a mark of zero for the assessment. Ensure you follow the specific assessment instructions in the section above.	✓
Level 3	**Compulsory:** **You must use GenerativeAI to complete your assessment** ***See assessment instruction for details*** This assessment fully integrates Generative AI, allowing you to harness the technology's full potential in collaboration with your own expertise. Always check your assessment instructions carefully as there may still be limitations on what constitutes acceptable use, and these may be specific to each assessment.	You will be **taught how** to use generative AI and assessed on its use. Your collaboration with GenerativeAI **must be clearly referenced** just as you would reference any other resource type used. **Click on the link below to learn how to reference GenerativeAI.** **https://library.kaplan.edu.au/referencing-other-sources/referencing-other-sources-generative-ai** In addition, you **must** include an appendix that documents your GenerativeAI collaboration including all prompts and responses used for the assessment. Unapproved use of generative AI as per assessment details during the content generation parts of your assessment may potentially result in penalties for academic misconduct, including but not limited to a mark of zero for the assessment. Ensure you follow the specific assessment instructions in the section above.	

Assessment Marking Guide

Marking Criteria ______| 40 marks	F (Fail) 0 – 49%	P (Pass) 50 – 64%	C (Credit) 65 – 74%	D (Distinction) 75 – 84%	HD (High Distinction) 85 – 100%
Code Functionality **______| 8 marks**	The program does not compile or execute correctly.	The program compiles and executes, but with limited functionality.	The program compiles and executes with expected functionality.	The program demonstrates additional functionality beyond the requirements.	The program exceeds all requirements and demonstrates exceptional functionality.
Class Implementation **______| 10 marks**	The classes are not implemented or do not adhere to the specifications.	The classes are implemented but lack proper inheritance or have limited functionality.	The classes are implemented correctly and inherit from the parent class.	The classes are implemented correctly with proper inheritance and demonstrate good design principles.	The classes are implemented flawlessly, showcasing exceptional design principles and code organisation.
Use of OOP Concepts **______| 10 marks**	There is no evidence of understanding or application of object-oriented programming concepts.	Basic object-oriented programming concepts are applied, but with limited understanding.	Object-oriented programming concepts such as encapsulation and inheritance are properly applied.	Object-oriented programming concepts are skilfully applied, resulting in code reusability and maintainability.	Object-oriented programming concepts are expertly applied, leading to highly reusable and maintainable code.
Code Style and Naming **______| 4 marks**	Code does not follow the PEP 8 conventions and lacks proper naming conventions.	Code follows some of the PEP 8 conventions but contains inconsistent or improper naming conventions.	Code follows most of the PEP 8 conventions and uses meaningful names for variables and functions.	Code follows PEP 8 conventions and uses consistent and meaningful names for variables and functions.	Code strictly adheres to PEP 8 conventions and uses clear and meaningful names for variables and functions.
Testing	No testing is performed on the classes.	Basic testing is performed to ensure minimal functionality.	Testing is performed on the classes to ensure correct functionality.	Thorough testing is performed to validate the classes' functionality and handle	Comprehensive testing is conducted, covering a wide range of scenarios and

_____| 8 marks				edge cases.	producing accurate results.
Feedback and grades will be released via MyKBS					